Digital Forensics and Analysis with Autopsy

Project Report

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1. Introduction

Purpose of the Project

The objective of this project is to understand the fundamentals of digital forensics by investigating a forensic disk image using the Autopsy forensic platform. The project focuses on identifying digital evidence, analyzing user activities, recovering artifacts, and answering forensic investigation questions provided in a Capture The Flag (CTF) challenge from TryHackMe.

The project provides practical experience in examining a compromised system while following a structured digital forensic investigation process.

Brief Overview of Digital Forensics and Autopsy

Digital forensics is the process of identifying, preserving, analyzing, and presenting digital evidence collected from computers, storage devices, and other digital media. It plays a vital role in cybercrime investigations, incident response, malware analysis, and legal investigations.

Autopsy is an open-source digital forensic platform built on The Sleuth Kit (TSK). It provides investigators with an easy-to-use graphical interface for analyzing disk images and extracting forensic artifacts such as:

- User accounts
- File metadata
- Browser history
- Deleted files
- Registry information
- Installed software
- System logs
- Network information
- Images and multimedia files

Autopsy automates many forensic tasks, making it one of the most widely used forensic investigation tools.

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2. Lab Setup

Tools Used

Tool	Purpose
TryHackMe	Digital Forensics Lab Environment
Autopsy	Digital Forensic Analysis Tool
Windows Virtual Machine (TryHackMe Lab)	Investigation Environment
Web Browser	Access TryHackMe Platform

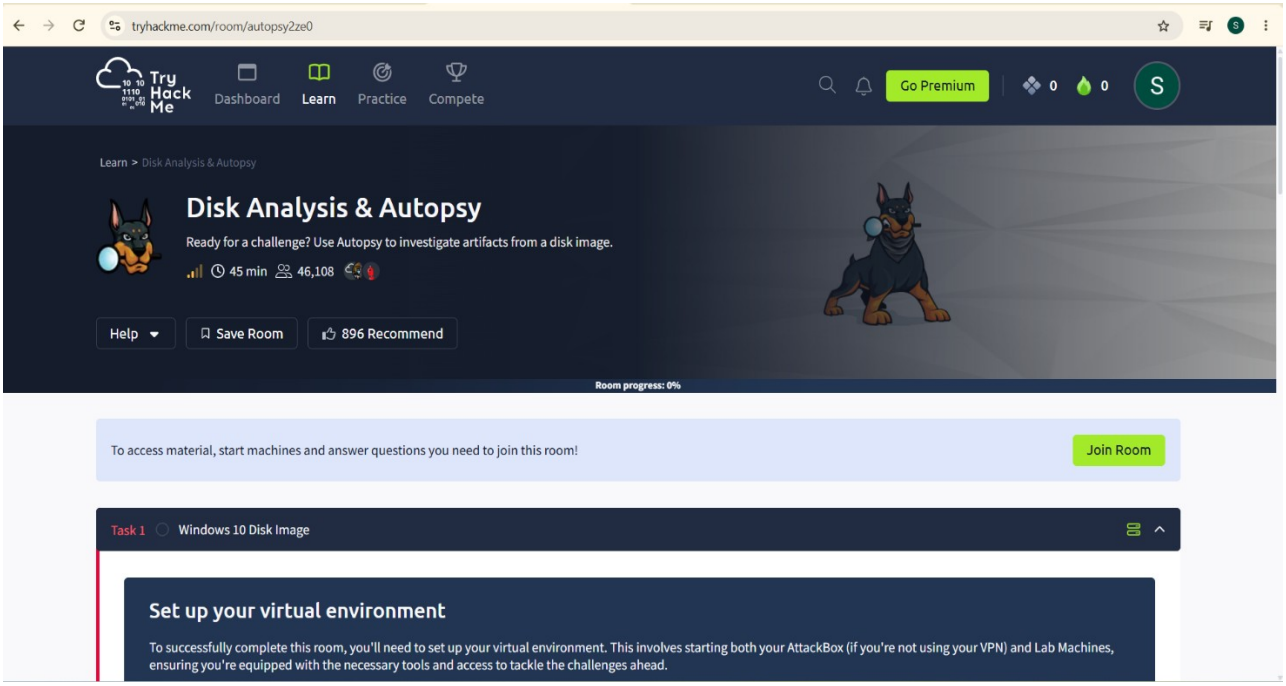
Installation / Lab Setup Process

The following steps were performed:

1. Logged into the TryHackMe platform.
2. Opened the **Disk Analysis and Autopsy** room.
3. Started the provided Lab Machine.
4. Launched the Autopsy forensic application.
5. Opened the provided forensic case.
6. Loaded the missing forensic image when prompted.
7. Allowed Autopsy to index and analyze the evidence.

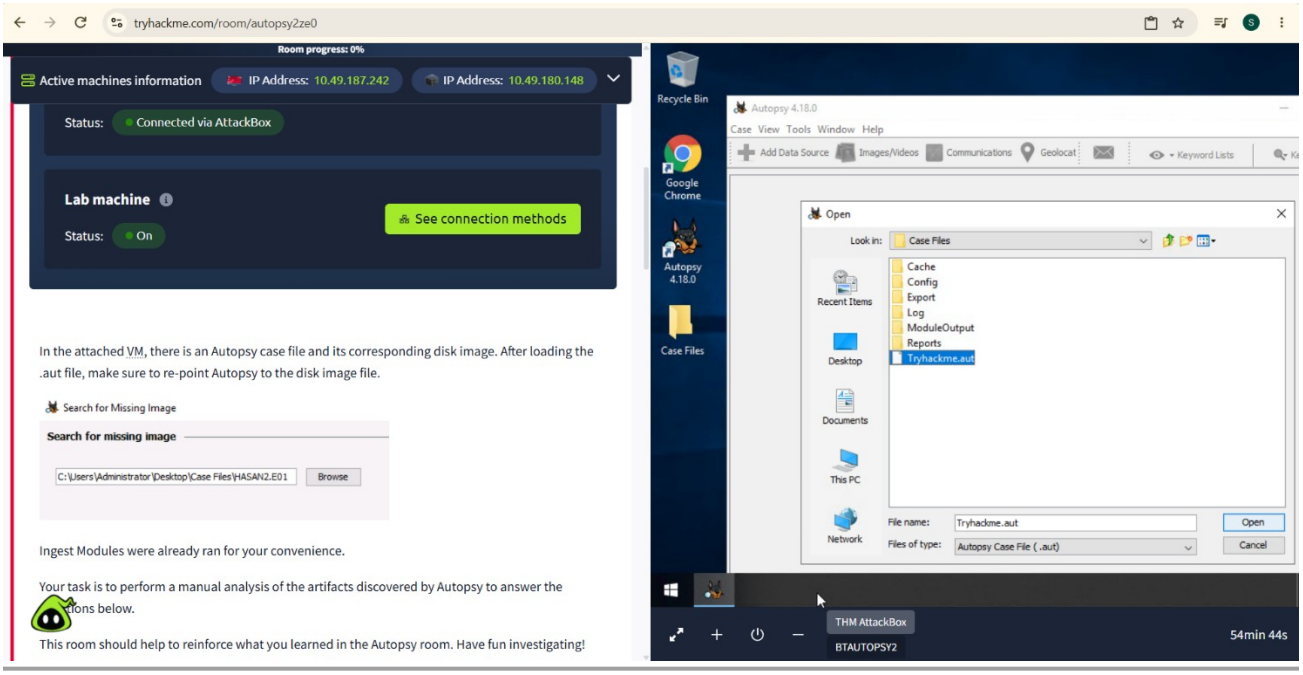
Screenshot 1

Working Environment



Screenshot 2

Autopsy Open with Loaded Case



3. Case Analysis

Description of the TryHackMe Challenge

The investigation involved analyzing a forensic disk image provided in the TryHackMe "Disk Analysis and Autopsy" room.

The objective was to recover digital evidence, identify user activities, examine system metadata, locate hidden artifacts, and answer multiple forensic questions based on the recovered information.

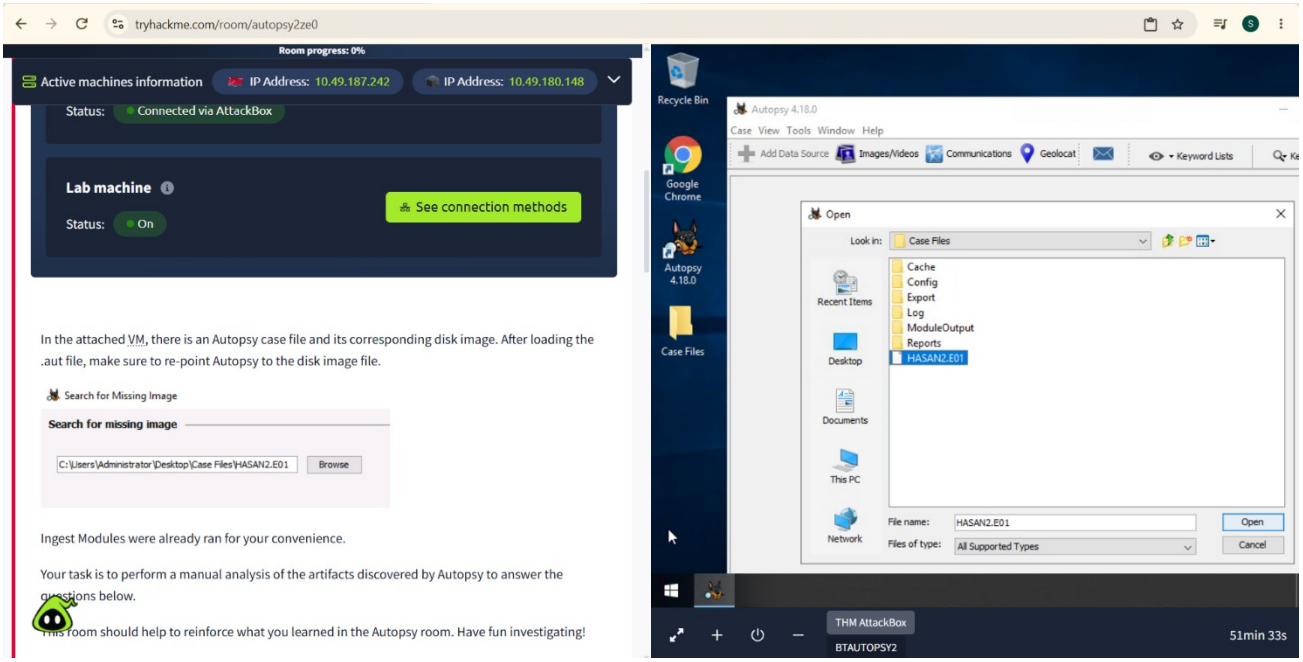
The challenge simulated a real-world forensic investigation where an examiner must analyze a Windows system image without modifying the original evidence.

Investigation Steps

Step 1 – Open the Case

- Launched Autopsy.
- Selected **Open Case**.
- Loaded the TryHackMe forensic case.

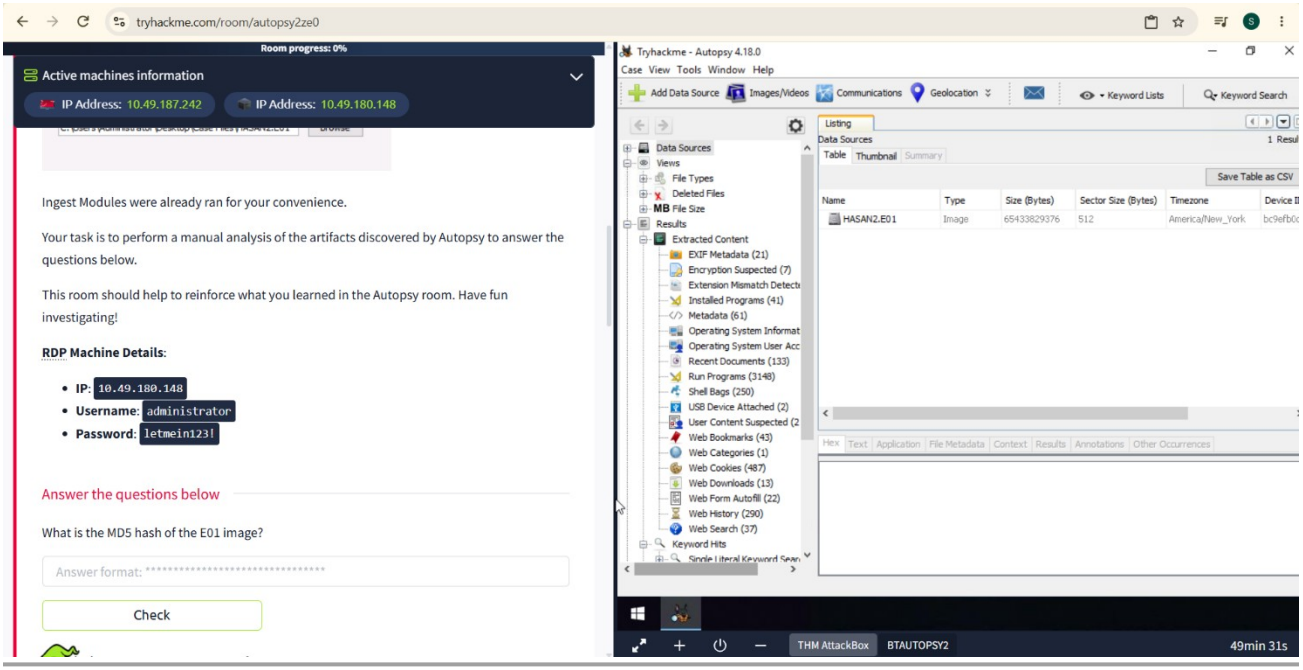
Screenshot



Step 2 – Load the Disk Image

- Selected the missing forensic image.
- Loaded the evidence into Autopsy.
- Allowed the indexing process to complete.

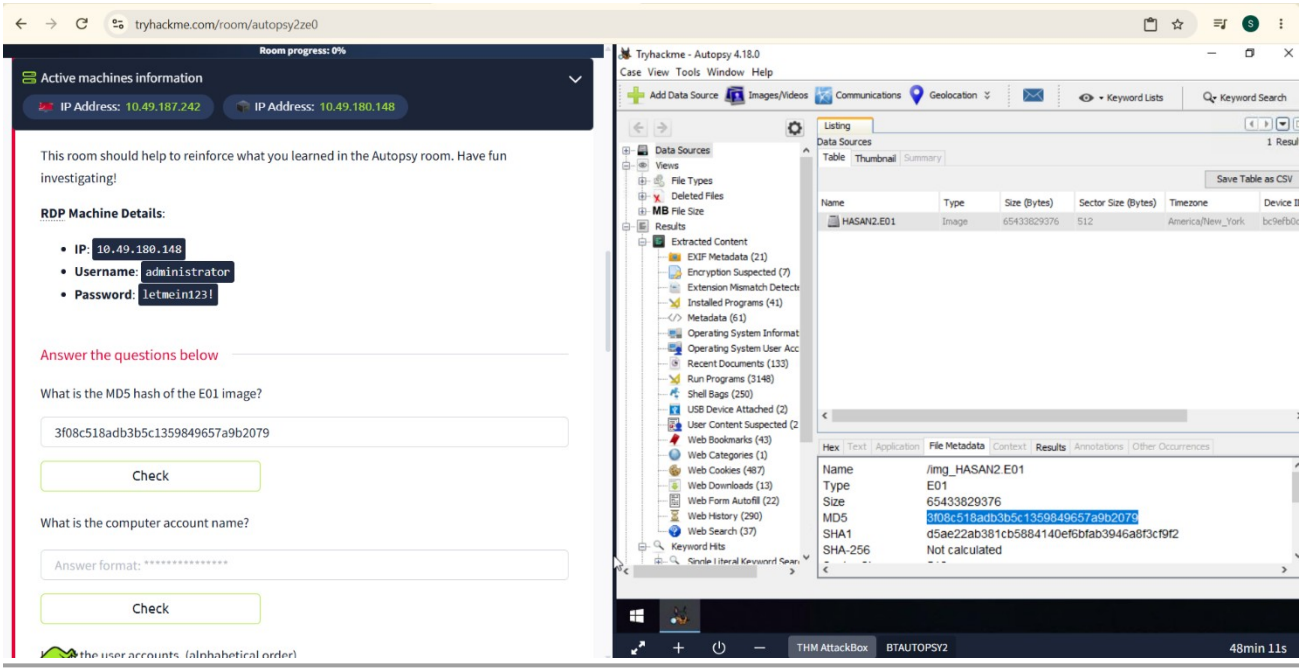
Screenshot



Step 3 – Verify Evidence Hash

- Opened image information.
- Retrieved the MD5 hash of the E01 forensic image.
- Verified the integrity of the evidence.

Screenshot



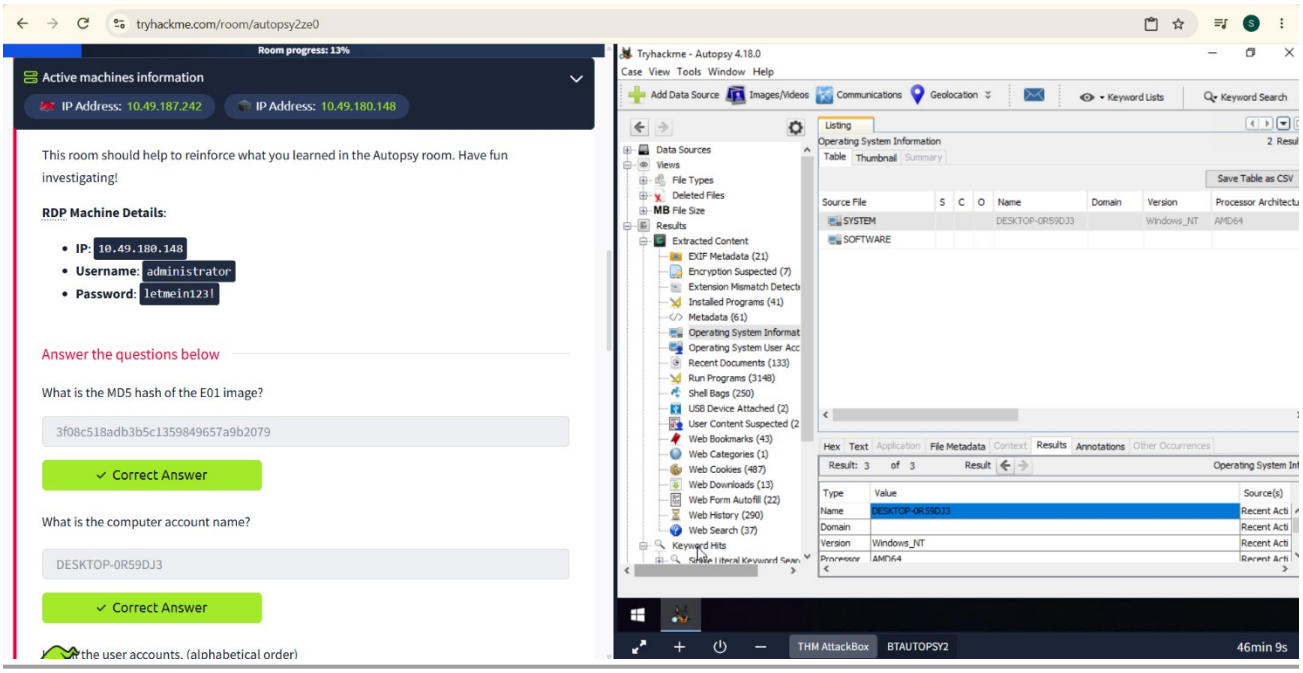
Step 4 – Analyze Operating System Information

Examined:

- Computer Name
- Operating System Information
- User Accounts
- Last Logged-in User

Recovered information was used to answer the corresponding forensic questions.

Screenshot



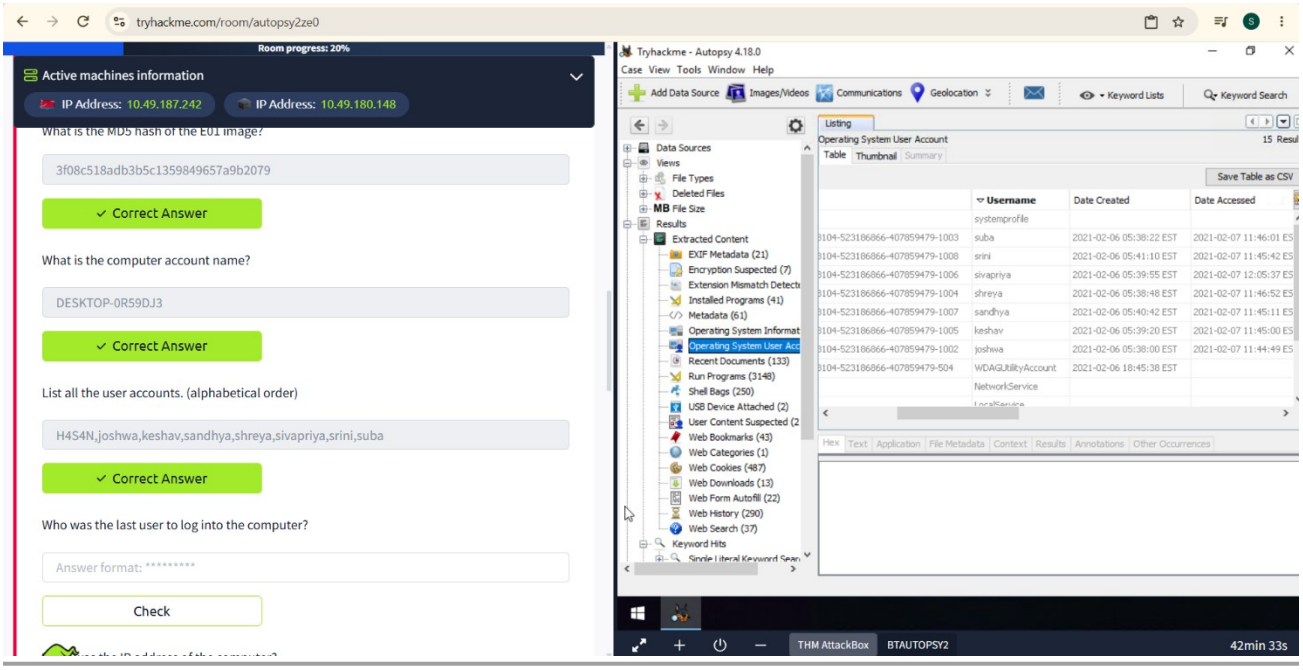
Step 5 – Investigate User Accounts

Reviewed:

- Available users
- Login timestamps
- User activity

Identified the last active user account.

Screenshot



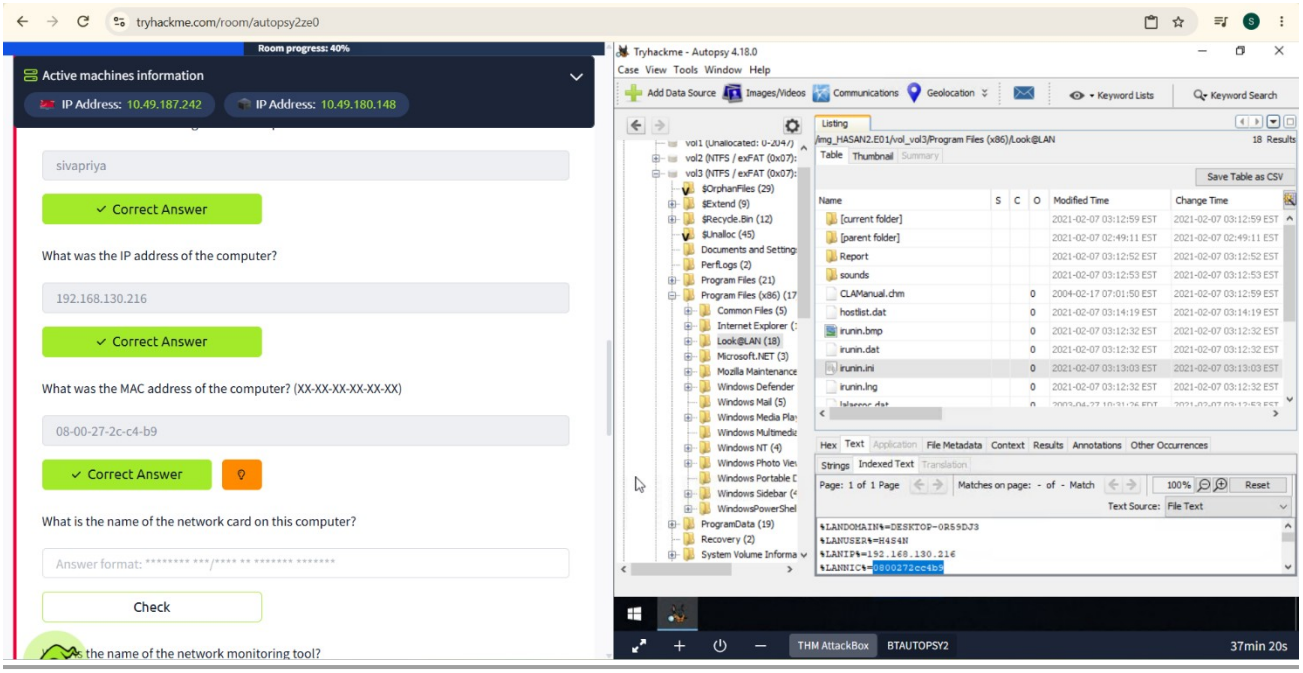
Step 6 – Analyze Network Information

Recovered:

- Computer IP Address
- Network Adapter
- Installed Network Monitoring Tool

Autopsy's file system browser and installed application data were used to identify the required information.

Screenshot



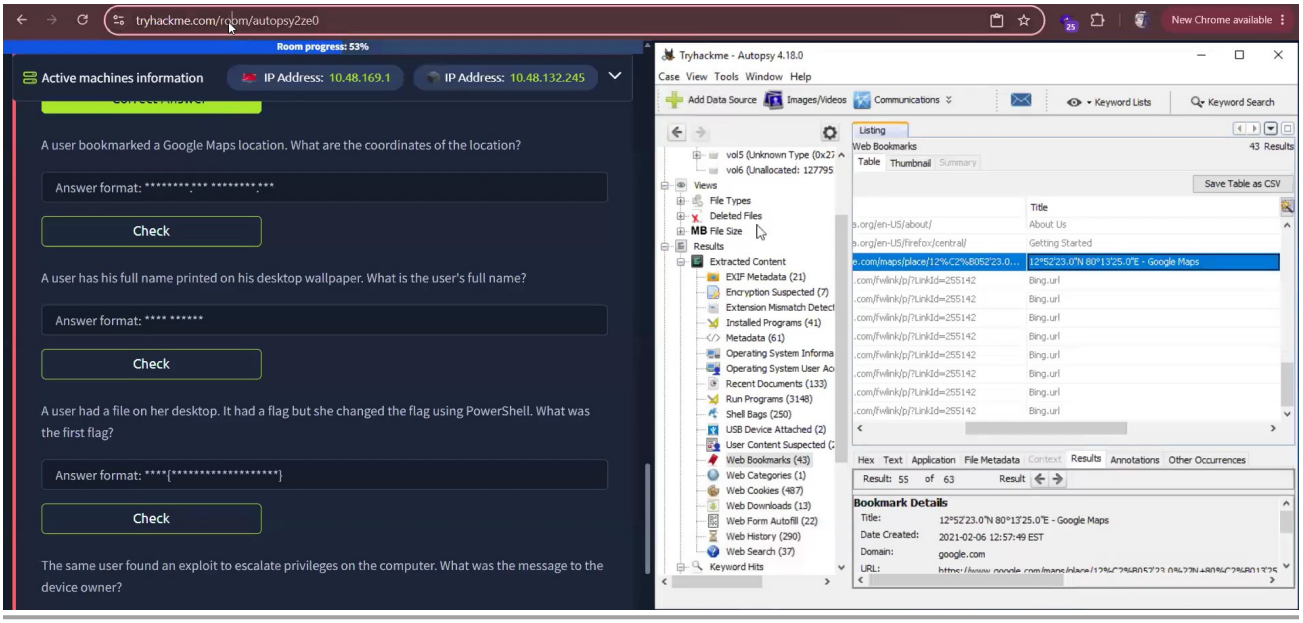
Step 7 – Examine Browser Artifacts

Opened Browser Bookmarks.

Recovered:

- Saved Google Maps bookmark
- Browser metadata

Screenshot



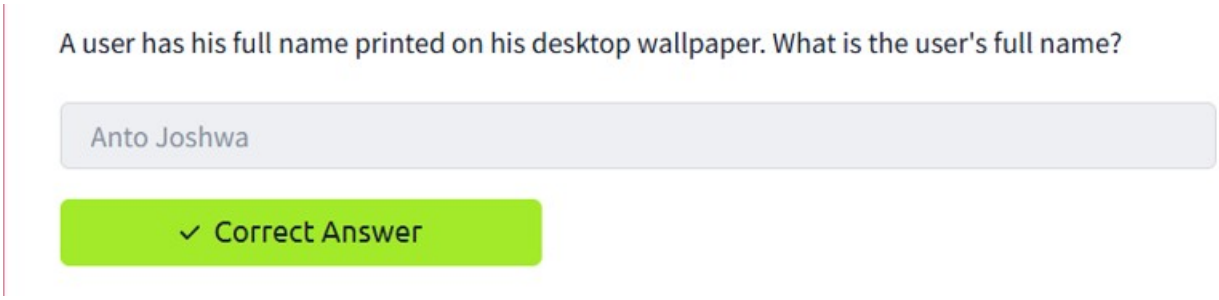
Step 8 – Examine Images

Navigated to:

- Images
- Videos

Recovered desktop wallpaper information and identified the user's full name.

Screenshot



Step 9 – Investigate PowerShell History

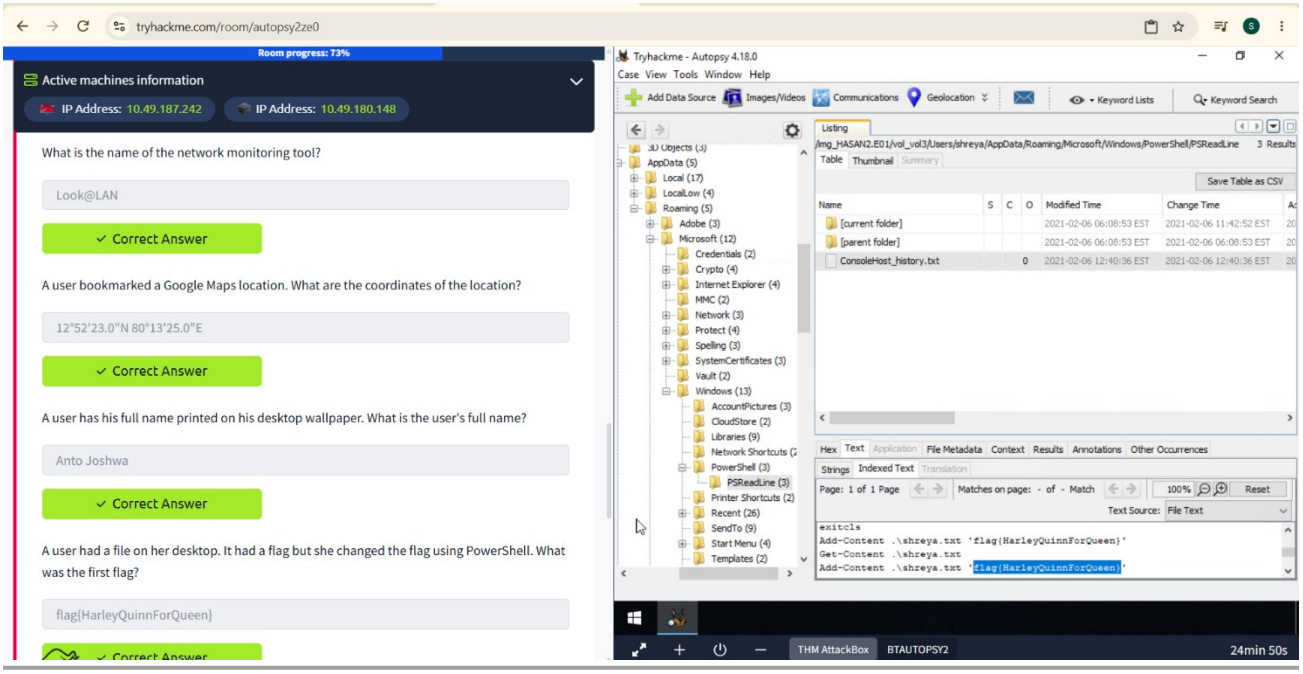
Browsed to:

Users
└── AppData
    └── Roaming
        └── Microsoft
            └── Windows
                └── PowerShell
                    └── PSReadLine

Recovered:

- Console History
- Previously executed commands
- Modified flag

Screenshot



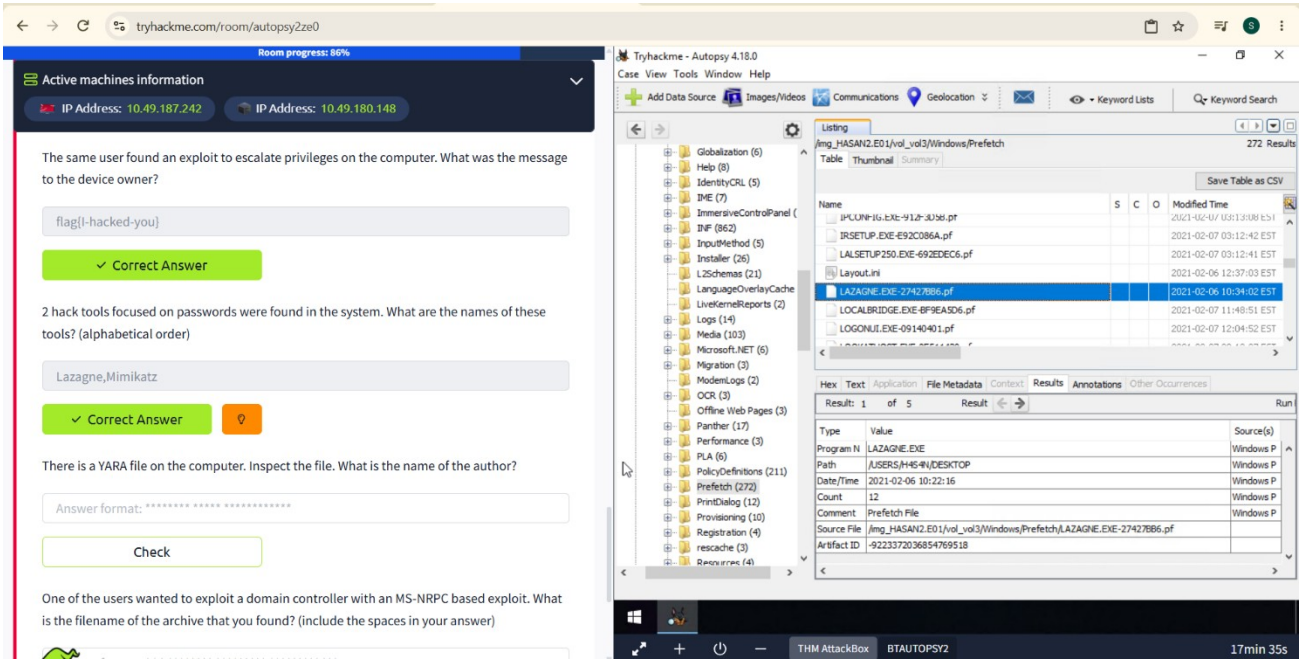
Step 10 – Investigate Desktop Files

Opened the user's Desktop directory.

Recovered:

- Text files
- Exploit-related documents
- Hidden flag

Screenshot



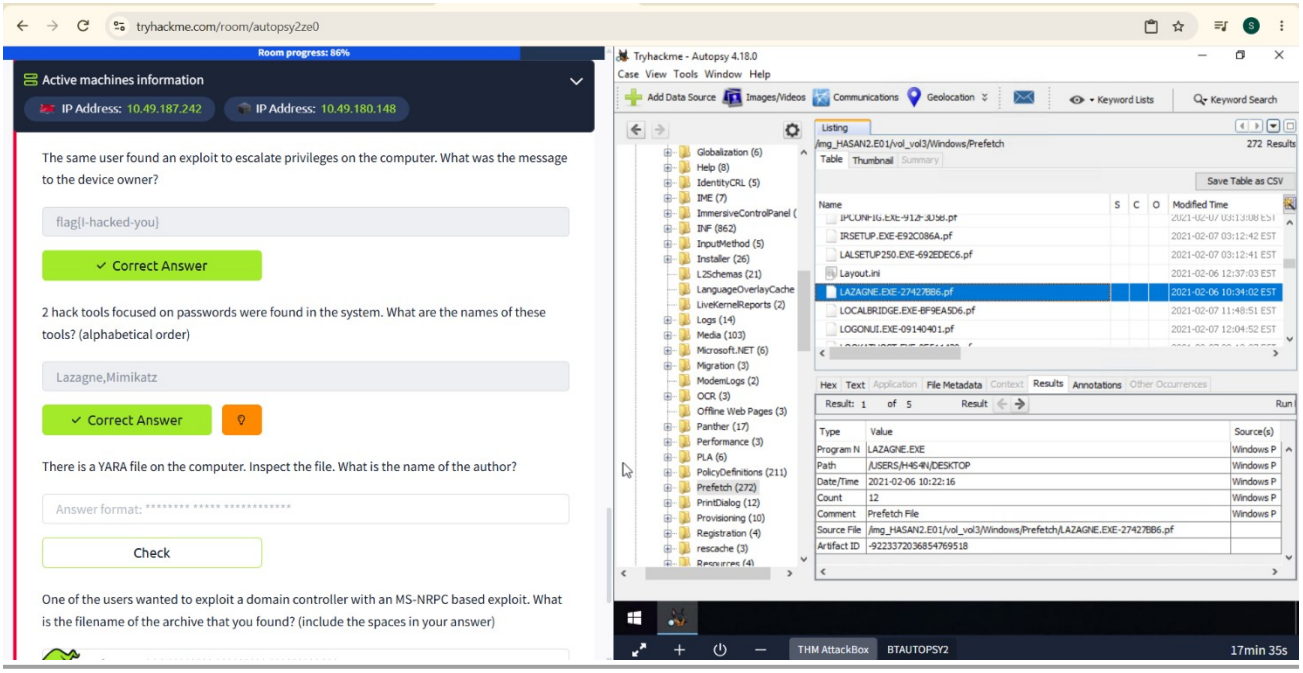
Step 11 – Examine Downloads

Analyzed downloaded files.

Recovered evidence included password-related hacking tools.

Also inspected Prefetch files to confirm execution.

Screenshot



Step 12 – Keyword Search

Performed a Keyword Search.

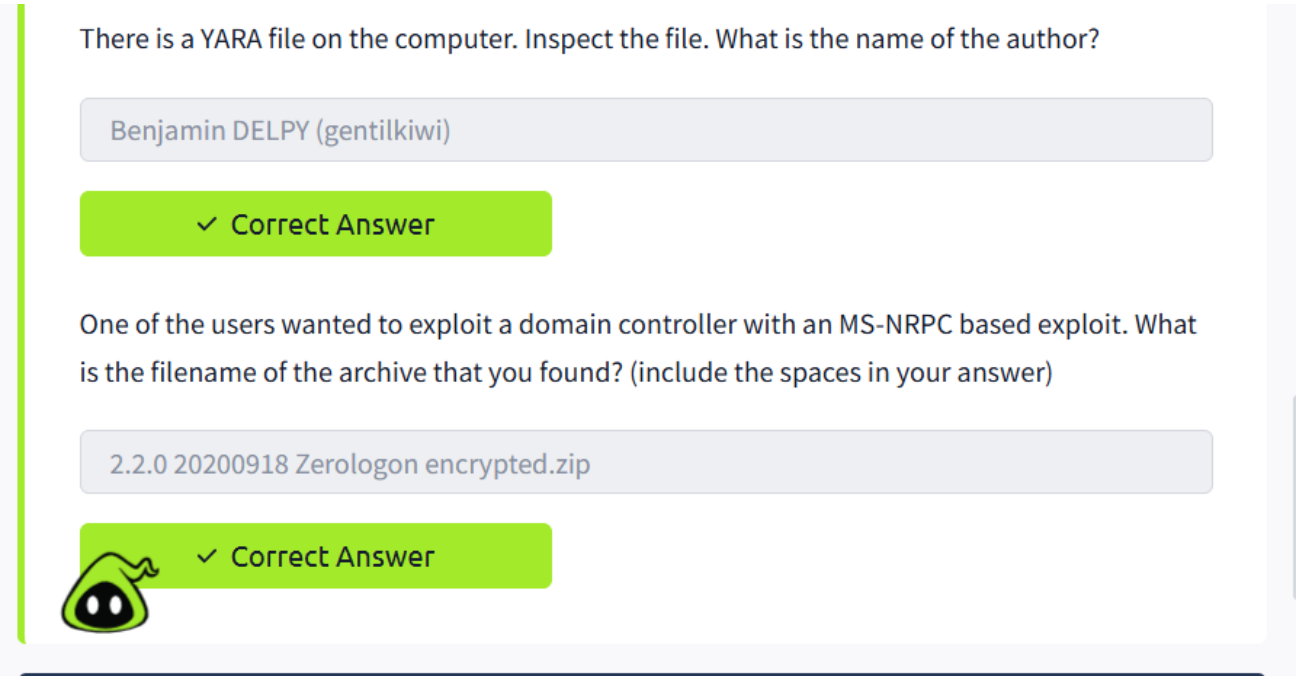
Keyword used:

yar

Recovered:

- YARA Rule
- Author Information

Screenshot



Step 13 – Recent Documents

Examined:

Users

→ AppData

→ Roaming

→ Microsoft

→ Windows

→ Recent

Recovered:

- Recently opened exploit file
- Exploit name

Recovered Data and Artifacts

The forensic investigation successfully recovered the following evidence:

- Computer account information
- User accounts
- Last logged-in user
- MD5 hash of forensic image

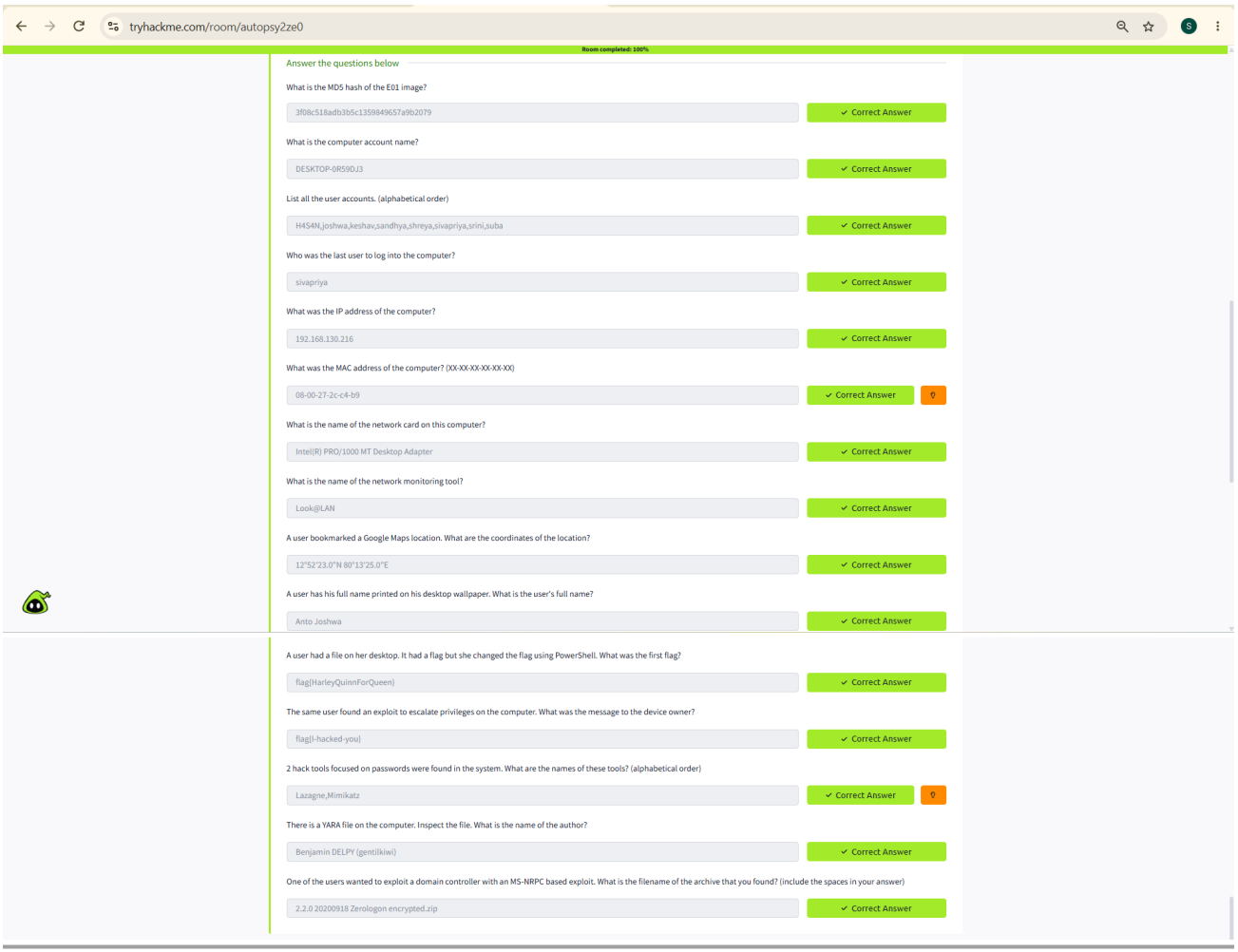
- IP address
- Network adapter information
- Installed network monitoring software
- Browser bookmarks
- Desktop wallpaper information
- PowerShell command history
- Desktop files
- Downloaded password recovery tools
- YARA rule file
- Recently accessed exploit files

Answers to CTF Questions

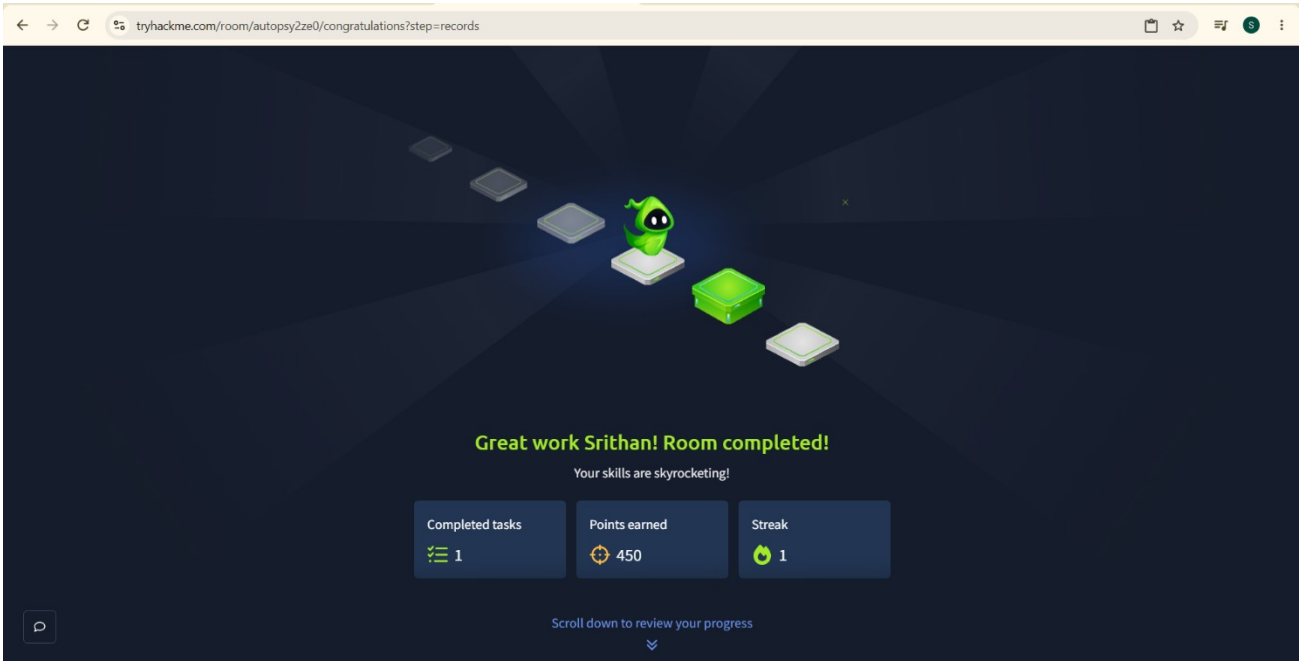
All forensic questions provided in the TryHackMe room were successfully answered by analyzing:

- Operating System Information
- User Accounts
- Installed Software
- Browser Artifacts
- PowerShell History
- Downloads Folder
- Images
- Recent Documents
- Keyword Search Results

Screenshot



Room Completion Screenshot



4. Key Findings

The forensic investigation revealed several important digital artifacts.

User Activity

- Multiple user accounts were identified.
- The last logged-in user was determined.
- Desktop activity was recovered.
- PowerShell command history revealed executed commands.

File System Evidence

Recovered:

- Desktop files
- Downloaded applications
- Recently opened documents
- Browser bookmarks

Metadata

Recovered metadata included:

- MD5 hash
- File paths
- User information
- System information
- Network configuration

Techniques Used

The investigation used several forensic techniques including:

- Disk Image Analysis
- File System Examination
- Metadata Analysis
- Browser Artifact Analysis

- PowerShell History Analysis
- Keyword Searching
- User Profile Analysis
- Recent Documents Analysis
- Installed Software Analysis

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5. Conclusions

Lessons Learned

This project provided practical experience in performing a digital forensic investigation using an industry-recognized forensic platform.

Key learning outcomes include:

- Understanding forensic workflows
- Preserving evidence integrity
- Recovering digital artifacts
- Examining user activities
- Identifying system metadata
- Using keyword searches effectively
- Investigating browser and PowerShell artifacts

Effectiveness of Autopsy

Autopsy proved to be an efficient forensic analysis platform because it:

- Supports multiple forensic image formats.
- Organizes evidence automatically.
- Displays user-friendly forensic artifacts.
- Simplifies metadata analysis.
- Enables rapid keyword searching.
- Integrates with The Sleuth Kit for detailed forensic examination.

Challenges Faced and Resolved

Challenge	Resolution
Missing forensic image during case loading	Loaded the required image from the Desktop Files directory.
Large number of forensic artifacts	Used categorized views in Autopsy for efficient navigation.
Locating specific evidence	Used Keyword Search and artifact filters.
Identifying recent user activity	Examined Recent Documents and PowerShell history.

6. References

1. TryHackMe – Disk Analysis and Autopsy Room *(Use the room URL you accessed)*
2. Autopsy Official Documentation
   https://www.autopsy.com/documentation/
3. The Sleuth Kit Documentation
   https://www.sleuthkit.org/
4. TryHackMe Official Website
   https://tryhackme.com/